

DECISION TREE (ID3 ALGORITHM) – ITERATION 1 TO ITERATION 3 NOTES

=====

1. DECISION TREE KYA HAI?

=====

Decision Tree ek Machine Learning Algorithm hai jo data ko step-by-step split karke prediction karta hai.

Example:

Input Features:

- Study
- Attendance
- Assignment
- Revision

Target:

- Pass (Yes/No)

Goal:

Data ko aise divide karna ki Pass aur Fail alag ho jaye.

=====

2. IMPORTANT TERMS

=====

Feature (Input):

Study, Attendance, Assignment, Revision

Target (Output):

Pass

Entropy:

Entropy batati hai data kitna mixed (Impure) hai.

Entropy = 0

→ Pure Data

→ Sab Pass ya Sab Fail

Entropy = 1

→ Maximum Confusion

→ Half Pass, Half Fail

Information Gain:

Information Gain batata hai kis feature se sabse zyada confusion kam hua.

Jis Feature ka Information Gain sabse zyada hoga,

Decision Tree usi feature ko split karega.

=====

ITERATION 1

=====

STEP 1

Pivot Table

Rows:

Study

Columns:

Pass

Values:

Count of Pass

Output:

Study	Pass No	Pass Yes	Total
No	12	12	24
Yes	3	23	26

STEP 2

Probability

Probability (Yes)

Formula

= Yes Count / Total

Excel

$$=C5/(B5+C5)$$

Example

$$12/24 = 0.5$$

Probability (No)

$$=B5/(B5+C5)$$

Example

$$12/24 = 0.5$$

STEP 3

Surprise

Formula

$$=-\text{LOG}(\text{Probability},2)$$

Example

$$\text{Probability} = 0.5$$

$$-\log_2(0.5)=1$$

STEP 4

Entropy

Formula

Entropy =

Probability(No) × Surprise(No)

+

Probability(Yes) × Surprise(Yes)

Excel

=F5*H5+E5*G5

Example

0.5×1

+

0.5×1

=

1

Entropy = 1

STEP 5

Weighted Entropy

Formula

$$\frac{=(\text{Entropy1} \times \text{Total1} + \text{Entropy2} \times \text{Total2})}{\text{Grand Total}}$$

Excel

$$\frac{=(I5*(B5+C5)+I6*(B6+C6))}{((B5+C5)+(B6+C6))}$$

Answer

0.748

=====

ITERATION 2

=====

Filter

Study = Yes

Remaining Records = 26

Pivot Table

Rows:

Attendance

Columns:

Pass

Output

Attendance	Pass No	Pass Yes	Total
No	3	9	12
Yes	0	14	14

Probability (Yes)

=C5/(B5+C5)

Probability (No)

=B5/(B5+C5)

Surprise

=-LOG(Probability,2)

Entropy

=Probability(No)×Surprise(No)

+

Probability(Yes)×Surprise(Yes)

Excel

=F5*H5+E5*G5

Weighted Entropy

=(Entropy1×Total1 + Entropy2×Total2)

/ Grand Total

Answer

0.374

=====

ITERATION 3

=====

Filter

Study = Yes

Attendance = No

Remaining Records = 12

Pivot Table

Rows:

Assignment

Columns:

Pass

Output

Assignment	Pass No	Pass Yes	Total
------------	---------	----------	-------

No	3	3	6
----	---	---	---

Yes	0	6	6
-----	---	---	---

Probability (Yes)

=C5/(B5+C5)

Probability (No)

=B5/(B5+C5)

Surprise

=-LOG(Probability,2)

Entropy

=Probability(No)×Surprise(No)

+

Probability(Yes)×Surprise(Yes)

Excel

=F5*H5+E5*G5

Weighted Entropy

=(Entropy1×Total1 + Entropy2×Total2)

/ Grand Total

Answer

0.5

=====

SAME FORMULAS FOR EVERY ITERATION

=====

Probability (Yes)

= Yes Count / Total

Probability (No)

= No Count / Total

Surprise

=-LOG(Probability,2)

Entropy

$=(\text{Probability}(\text{No}) \times \text{Surprise}(\text{No}))$

+

$(\text{Probability}(\text{Yes}) \times \text{Surprise}(\text{Yes}))$

Weighted Entropy

$=(\text{Entropy} \times \text{Number of Records})$

/ Total Records

=====

DECISION TREE FLOW

=====

Original Dataset

↓

Iteration 1

(Study)

↓

Iteration 2

(Attendance)

↓

Iteration 3
(Assignment)



Iteration 4



Iteration 5



...



Iteration 12



Leaf Node (Entropy = 0)

=====

INTERVIEW POINT

=====

Decision Tree har iteration me:

1. Best Feature choose karta hai.
2. Entropy calculate karta hai.
3. Information Gain calculate karta hai.
4. Best Split select karta hai.
5. Ye process recursively repeat hoti hai jab tak data pure (Entropy = 0) na ho jaye ya stopping condition na aa jaye.